1. **The Secret to Longevity**. Scientists are interested in whether the energy costs involved in reproduction affect longevity[1]. In this experiment, 125 male fruit flies were divided at random into five sets of 25. In group 1, the males were kept by themselves. In groups 3 and 5, the males were supplied with one or eight receptive virgin female fruit flies per day, respectively. In groups 2 and 4, the males were supplied with one or eight unreceptive (pregnant) female fruit flies per day, respectively. Other than the number and type of companions, the males were treated identically. The longevity of the flies was observed.

   You can access the data as a csv file at: https://stat158.berkeley.edu/spring-2026/data/fruit-flies/fruit-flies.csv

   a. Visualize the data and interpret what you see. Be sure to comment both on the center of the distributions and their spread.

   b. *ANOVA "by hand"*. Without using any of the built-in commands (`aov()`, `anova()`) create a one-way ANOVA table minus the p-values. You're welcome to use either a base R or `dplyr` approach. It is a bit tedious, however it's useful to cobble together each of the statistics in an ANOVA table just once in your lives. You can check your answers with the output of `aov()` shown below.

```
                Df Sum Sq Mean Sq F value Pr(>F)
factor(trt)    4  11939  2984.8   13.61
Residuals    120  26314   219.3
```

   c. State your conclusion about the scientific question using Model-based inference. Be clear about the data generation model your approach relies upon and the null hypothesis that you're evaluating. To calculate the p-value, use the `pf()` function and then verify it using the `aov()` function[2].

   d. State your conclusion about the scientific question using Randomization-based Inference. Be clear about the data generation model your approach relies upon and the null hypothesis that you're evaluating. For your test statistic, use the F-statistic that you built the code for from part a above.

   e. How do the results of the model-based and randomization-based inference compare? Do they lead to the same conclusion? Do they rely on the same assumptions? Which do you think is more appropriate for this problem?

2. **More Comparisons for Longevity**. In the experiment regarding the fruit flies, the ANOVA analysis only tests whether there were any differences between the group means. Suppose there was further interest in whether the 1 pregnant group was different from the 1 virgin group (for questions a and b, restrict the data set to these two groups).

   a. Perform a randomization test to compare these two groups using the t-statistic. Assume the two groups have the same variance (revisit your notes on the t-test and the pooled estimate of the SE).

[1] Question from Problem 3.2 in Oehlert textbook, original data from Hanley and Shapiro (1994), ``Sexual activity and the lifespan of male fruitflies: a dataset that gets attention," *Journal of Statistics Education 2*.

[2] Recall all named probability distributions have a CDF in R with the prefix `p`. The F-distribution is no exception, so you can use `pf()` to calculate the p-value for your ANOVA table. The arguments to `pf()` are the F-statistic, the degrees of freedom for the numerator, and the degrees of freedom for the denominator. Recall a test using the F statistic is a right-tailed test. The standard anova output is available with `summary(aov(y ~ x, data))` (be sure your `x` is a factor vector).

b. Repeat the randomization test but this time use an F-statistic. How do the results of this analysis this compare to the results in part (a)?

c. Calculate an ANOVA table for this comparison and report the value of $\sqrt{MS_{residuals}}$, which is an estimate of σ. How does that compare to the estimate of the standard deviation that you used for the t-statistic (s_p)?

You are welcome to use built-in R commands for this question, if appropriate.

More questions to come next week!